Alben Rome B. Bagabaldo

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Personal Website ◇ Google Scholar ◇ LinkedIn ◇ GitHub

RESEARCH INTERESTS

Methodology: Data Science, GIS, Spatial AI, Network Science, Time Series Modeling, Knowledge Graphs, Machine Learning, Digital Twins, Simulation, Optimization, Uncertainty Quantification, Data-Driven Decision-Making
Applications: Transportation Planning, Public Transit, Human Mobility, Transportation Accessibility and Reliability, Climate and Infrastructure Resilience, Urban Analytics, Emerging Mobility Technologies

EDUCATION

University of California, Berkeley — Berkeley, CA
Ph.D., Civil and Environmental Engineering — 2024
Labs: Mobile Sensing Lab / Berkeley AI Research (BAIR) Lab & HumNet Lab;
Dissertation: "Navigating Urban Traffic: From Data to Simulations to Real-World Impacts".
Adviser: Prof. Alexandre M. Bayen — Committee Members: Prof. Marta C. Gonzalez, Prof. Joan Walker.

Mapúa University — Manila, Philippines
M.S., Civil Engineering, Transportation Engineering — 2016
Thesis: "Determination of Barriers towards Addressing Mobility and Accessibility through Traffic Simulation: Case of Intramuros, Manila".
Adviser: Dr. Francis Aldrine Uy

Southern Luzon State University — Quezon, Philippines
B.S., Civil Engineering, *cum laude* — 2012

ACADEMIC APPOINTMENTS

Rutgers University – New Brunswick — New Brunswick/Piscataway, NJ
Postdoctoral Associate, Center for Advanced Infrastructure and Transportation; with Prof. Hao Wang — 2025–present

- Developing geographic data science methods for transit resilience and climate-risk analysis. Current work integrates GTFS service data, road networks, flood hazard layers, accessibility/POI data, and energy-operational constraints to evaluate garage-level vulnerability and service continuity for NJ TRANSIT's battery-electric bus transition; translate spatial evidence into planning and decision-support outputs.

Princeton University — Princeton, NJ
Postdoctoral Research Associate, Complex Infrastructure Systems Group; with Asst. Prof. Jürgen Hackl — 2024–2025

- Led interdisciplinary research on urban digital twins for intelligent intersections, combining spatial data, probabilistic modeling, Monte Carlo simulation, network analysis, and spatial AI approaches to study mobility, safety, and resilience; collaborated with Rutgers and the Municipality of Princeton; co-developed competitive proposals.
- Professional development: GradFutures Sustainability Learning Cohort.

University of California, Berkeley — Berkeley, CA
Graduate Student Instructor for multiple courses; Reader/Grader for IE 242 – Applications in Data Analysis — 2019–2024

- Led sections, supported curriculum delivery, developed instructional materials, graded assignments/exams, and mentored students in transportation, spatial analytics, data science, and energy systems courses.

Mapúa University — Manila, Philippines
Instructor/Lecturer; Program Chair; Senior Science Research Specialist; Research Associate — 2014–2019

- *Instructor-of-Record* for undergraduate civil engineering courses; *Program Chair* for Construction Engineering and Management (Aug. 2018-Jan. 2019); GIS specialist for DOST-funded Mapúa Phil-LiDAR 1 flood-hazard mapping; supported desktop GIS workflows, mentored local government units, and coordinated official turnover of hazard maps.

Southern Luzon State University — Quezon, Philippines
Contract of Service Instructor, College of Engineering — 2013–2014

COURSES TAUGHT

University of California, Berkeley – Graduate Student Instructor

Graduate-level

- CE C263H/CP C257H – **Human Mobility and Network Science** (Fall 2023): urban and global mobility, spatial interactions with built and natural environments, and network science approaches to mobility systems. Class size: ∼60 students.
- CE 295 – **Data Science for Energy** (Spring 2023, Fall 2021): energy system management and control, batteries, EVs, renewables, power systems, smart buildings, modeling, state-space, observers, feedback control, and optimization. Class size: ∼60 students.
- CE 263N/CP 257 – **Scalable Spatial Analytics / Data Science for Human Mobility and Socio-technical Systems** (Fall 2022): spatial statistics, exploratory data analysis, spatial data mining, machine learning, and spatial data visualization. Class size: ∼60 students.
- CE 259 – **Public Transportation Systems** (Spring 2022): transit technology, operations, management, policy, and financing. Class size: ∼35 students.

Undergraduate

- PHYSICS 8A – **Introductory Physics** (Spring 2024, Summer 2023): forces, kinetics, equilibria, fluids, waves, and heat. Class size: ∼25 students per section.
- CE C88/CP C88 – **Data Science for Smart Cities** (Spring 2021): transportation, power-grid, communication, crowd-sensing, and remote-sensing data for urban services, spatial data, and smart-city analytics. Class size: ∼30 students.

Mapúa University – Instructor of Record

- Undergraduate courses in Methods of Research, Transportation Engineering, and Engineering Economy; supervised undergraduate theses.

Southern Luzon State University – Instructor of Record

- Undergraduate courses in Traffic Engineering, Soil Mechanics, and Fluid Mechanics.

PUBLICATIONS

Google Scholar: 68 total citations; h-index: 4; i10-index: 2 (as of October 29, 2026).

Peer-Reviewed Journal Articles

J1. **Bagabaldo A.R.** & H. Wang. (Under review, 2026). Resilience of Battery-Electric Bus System to Natural Hazards: Challenges, Strategies and Policy. Manuscript under review at *Transport Policy*.

J2. **Bagabaldo, A. R.**, & Hackl, J. (Accepted, October 2026). Digital twins for intelligent intersections: A literature review. *Transportation Research Interdisciplinary Perspectives*. *arXiv preprint* arXiv:2510.05374. [link] *[GS: 6 citations]*

J3. Tang, Y., Alhadlaq, A., **Bagabaldo, A. R.**, & Gonzalez, M. C. (2025). Designing transit routes based on vehicle routing behavior determined through location-based services data. *EPJ Data Science*, *14*, Article 45. [link] *[GS: 5 citations]*

J4. **Bagabaldo, A. R.**, Gan, Q., Bayen, A. M., & Gonzalez, M. C. (2024). Impact of navigation apps on congestion and spread dynamics on a transportation network. *Data Science for Transportation*, *6*, Article 12. [link] *[GS: 14 citations]*

J5. Chou, F.-C., **Bagabaldo, A. R.**, & Bayen, A. M. (2022). The lord of the ring road: A review and evaluation of autonomous control policies for traffic in a ring road. *ACM Transactions on Cyber-Physical Systems*, *6*(1), 1–25. [link] *[GS: 32 citations]*

J6. **Bagabaldo, A. R.**, Añosa, J., Gapusan, J., Soriano, K., Uy, F., & Bacero, R. (2017). Determining the potential effects of the proposed removal of south provincial buses along EDSA using commuters' perception. *Journal of the Eastern Asia Society for Transportation Studies*, *12*, 489–504. [link] *[GS: 1 citation]*

Refereed Conference Proceedings

C1. **Bagabaldo, A. R.**, & Hackl, J. (2025). Improving pedestrian safety at intersections using probabilistic models and Monte Carlo simulations. In *Proceedings of the International Conference on Transportation and Development (ICTD 2025)*. Preprint: arXiv:2503.07805. *[GS: 4 citations]*

C2. **Bagabaldo, A. R.**, & Gonzalez, M. C. (2022). Predicting traffic flow on faulty traffic detectors using machine learning techniques. In *Proceedings of the International Conference on Transportation and Development 2022*. [link] *[GS: 2 citations]*

In Preparation

IP1. **Bagabaldo, A.R.** & H. Wang. Impact of Flooding on Battery-Electric Bus Operations and Energy Demand.

IP2. **Bagabaldo, A.R.** & H. Wang. Multi-Criteria Vulnerability Assessment for Battery-Electric Buses.

IP3. S. Jenks, G. Weissman, H. Wang, **Bagabaldo, A.R.**, W. Berry, & M. Tuozzolo. An Optimization Framework for Evaluating Energy Cost Reduction Strategies for Transit Electric Bus Fleets.

Preprint

PP1. C. Zhang, T. LaRock, **Bagabaldo, A.R.**, & J. Hackl. Rethinking the Sioux Falls Network: Insights from Path-Driven Higher-Order Network Analysis. *arXiv preprint* arXiv:2508.06234. [link].

CONFERENCE PRESENTATIONS / INVITED TALKS

Conference Presentations

P1. **Bagabaldo, A. R.**, & Wang, H. (2026, June). Assessing and Mitigating Natural Hazard Impacts on Battery-Electric Bus Operations: A Garage-Level Resilience Index Approach. Oral presentation at the 10th International Symposium on the Use of Public Transit Automated Data for Planning, Operations, and Management.

P2. **Bagabaldo, A. R.**, & Hackl, J. (2025, October). Digital Twins for Intelligent Intersections: Merging Real-World Data, Probabilistic Modeling, and Monte Carlo Simulations to Improve Safety. Online oral presentation at the DigiTwin Conference 2025: DT for Smart Construction and Cities.

P3. **Bagabaldo, A. R.**, Gan, Q., Bayen, A. M., & Gonzalez, M. C. (2023, June). Price of information and the spread of congestion. Oral presentation at the ASCE International Conference on Transportation and Development.

P4. **Bagabaldo, A. R.**, Gan, Q., Lee, J., Bayen, A. M., & Gonzalez, M. C. (2023, June). Large-scale simulation-based understanding of selfishness in routing. Poster presented at the ASCE International Conference on Transportation and Development.

P5. **Bagabaldo, A. R.** (2023, June). Research highlights for PhD students. Panel speaker at the ASCE International Conference on Transportation and Development.

P6. Cabannes, T., **Bagabaldo, A. R.**, Lee, J., Gan, Q., Jain, A., Blondel, A., & Bayen, A. M. (2023, January). Creating, calibrating, and validating large-scale microscopic traffic simulation. *Transportation Research Board 102nd Annual Meeting*. TRID: 2087449. *[GS: 4 citations]*

Invited Talks / Guest Lectures

T1. Bagabaldo, A. R. (2024, April 17). Guest lecture in CE 295 – Data Science for Energy: A review and evaluation of autonomous control strategies for traffic in a ring road. University of California, Berkeley.

T2. Bagabaldo, A. R. (2017, May 31). Media interview – *The Service Road by James Deakin*. CNN Philippines. [link]

T3. Bagabaldo, A. R. (2017, March 16). Mapúa Phil-LiDAR 1 Maps Turnover Ceremony: Big Data, LiDAR, and flood-hazard mapping. Mapúa University.

T4. Bagabaldo, A. R. (2017, March 2). Geospatial information and civil systems for sustainable and inclusive transportation. Mapúa University.

T5. Bagabaldo, A. R. (2017, February 17). Presenter, Intelligent Transportation Systems Forum. De La Salle University–Manila.

AWARDS & FELLOWSHIPS

- **Outstanding Graduate Student Instructor Award**, City and Regional Planning, UC Berkeley – campus-wide recognition, approximately top 10% of GSIs. List of Recipients, AY 2023–2024. 2024
- **Conference Travel Grant Awards**, UC Berkeley Graduate Division. 2022
- **Ph.D. Overseas Scholarship**, Commission on Higher Education – Philippine California Advanced Research Institutes (CHED–PCARI). 2019–2023

- **Dean's Council Scholarship**, Mapúa University. 2016
- **JSCE Study Tour Grant**, Japan Society of Civil Engineers. 2016
- **Young Future Energy Leaders**, Masdar Institute of Science and Technology, Abu Dhabi, UAE. 2016

MENTORSHIP & SERVICE

Research Mentorship

University of California, Berkeley

- *Undergraduate:* Arya Bakhtiar (EECS, 2023–2024; now M.S. student at Stanford), Preston Fu (EECS, 2022–2023, incoming PhD students at University of Washington), Qianxin Gan (EECS, 2022–2023; now Software Engineer at Google), John Lee (EECS, 2021–2022; now Software Engineer at Wing).
- *Graduate:* Yuhan Tang (CEE Systems, 2023–2024; now PhD student at MIT).

Princeton University

- *Graduate:* Chen Zhang and Esteban Nocet-Binois (CEE Ph.D. students).

Leadership

- Vice President & Founding Member,
  Representation of Asian and Pacific Islanders in CEE (RAPID-CEE), UC Berkeley. 2023–2024
- Faculty Adviser, Transportation Science Society of the Philippines – Mapúa Student Chapter. 2016–2018
- Vice President, College of Engineering Student Council, Southern Luzon State University. 2010–2011

Referee Service

Journals

- IEEE Transactions on Intelligent Transportation Systems
- IET Intelligent Transport Systems
- Engineering Applications of Artificial Intelligence

Conferences

- Transportation Research Board Annual Meeting
- IEEE Intelligent Transportation Systems Conference
- IEEE Intelligent Vehicles Symposium
- International Symposium on Automation and Robotics in Construction

PROFESSIONAL MEMBERSHIPS

- Eastern Asia Society for Transportation Studies – Transportation Science Society of the Philippines: Regular Member, 2017–present; Associate Member, 2016–2017.
- Japan Society of Civil Engineers: Associate Member, 2016–present.
- American Society of Civil Engineers: Affiliate Member, 2024–present; Student Member, 2022–2024.

REFERENCES

Prof. Alexandre M. Bayen
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